# Structured Schemas for LLM-Modeler Collaboration in Quantitative Systems Pharmacology Model Calibration

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**Abstract**

Quantitative systems pharmacology (QSP) models require calibration data from published literature, yet manual curation produces inconsistent documentation while large language model (LLM) extraction exhibits hallucination and fabrication errors unacceptable for quantitative modeling. We present MAPLE (Model-Aware Parameterization from Literature Evidence), a framework that uses structured validation schemas as a collaboration interface between LLMs and modelers. Two complementary schemas capture calibration data at different scales: one for isolated experiments that constrain individual parameters through simplified forward models, and one for clinical and in vivo endpoints that constrain the full model through species-level observables. Both schemas separate data extraction from modeling decisions, capturing literature values with full provenance in a machine-verifiable form. Targeted validators catch characteristic

LLM errors: value-in-snippet matching detects hallucinated values, DOI resolution flags fabricated citations, and code execution catches malformed forward models. We evaluate MAPLE on 87 calibration targets for a pancreatic ductal adenocarcinoma (PDAC) QSP model, using two collaboration modes: batch LLM extraction followed by interactive curation, and interactive extraction where modeler and LLM collaborate in real time. Both modes required substantial modeler input: the modeler changed forward model types in 65% of SubmodelTargets, adjusted prior parameters in 46%, and revised source relevance assessments in all files. Interactively extracted targets embedded modeler effort in the extraction process, producing near-final output. The schemas ensure completeness and enable reproducible, provenance-rich calibration regardless of workflow.

# 1 Introduction

Quantitative systems pharmacology models translate mechanistic understanding of disease biology into predictions of drug response [1]. QSP models can comprise tens to hundreds of ordinary differential equations, and calibration requires data from diverse sources including targeted experimental studies and clinical endpoints [2, 3]. Because virtual population generation and virtual clinical trials depend on parameter distributions rather than point estimates [4, 5], calibration must quantify uncertainty. As QSP increasingly integrates with machine learning [6, 7, 8, 9, 10], the bottleneck of manual parameter extraction becomes more acute.

Extracting calibration data from literature requires judgment about applicability, uncertainty, and potential biases, with full provenance for reproducibility. We use the term *calibration target* to refer to a structured representation of published data that captures what was measured, under what conditions, with what uncertainty, and how that measurement relates to model parameters. Some studies isolate specific mechanisms (e.g., in vitro proliferation assays, ex vivo cytotoxicity studies) and directly constrain individual parameters, while clinical and in vivo endpoints constrain the integrated model.

Manual curation remains the standard approach but is labor-intensive, with inconsistent documentation across individuals and projects. When personnel change, the reasoning behind parameter choices may be lost. Without structured representations, combining constraints from multiple experiments requires ad-hoc scripting with no standard path from extracted data to inference code.

Traditional NLP pipelines for automated extraction require task-specific training data [11, 12, 13] and do not generalize to the diverse parameter types in QSP modeling.

Large language models offer a more flexible approach. LLMs can identify relevant literature given mechanistic context, read scientific text, and extract information into structured templates [14, 15, 16, 17, 18]. However, LLMs exhibit hallucination and fabrication errors that are unacceptable for quantitative modeling [19, 20, 21]. Many of these failure modes, however, are amenable to systematic detection: hallucinated values can be caught by requiring verbatim source text, fabricated citations through DOI resolution, and internal inconsistencies through schema validation. This suggests a workflow where LLMs handle extraction while automated validators catch characteristic errors, with Pydantic [22], a Python data validation library, enabling a self-correcting retry loop.

We present MAPLE (Model-Aware Parameterization from Literature Evidence), a framework for LLM-assisted QSP model calibration with four components: (1) model-aware literature search using mechanistic context to guide LLM web search; (2) validated YAML schemas for two calibration regimes (isolated experiments constraining individual parameters, and clinical/in vivo endpoints constraining the full model), separating data extraction from model specification while preserving traceability; (3) targeted validators for hallucination detection (value-in-snippet matching), citation verification (DOI resolution), and code correctness; and (4) code generation for Bayesian inference. We evaluate MAPLE on calibration targets for a PDAC QSP model and report metrics on validation performance and extraction reliability.

# 2 Methods

## 2.1 Model-Aware Literature Search

The user specifies which model parameters need calibration, and the system builds a prompt that guides an LLM with web search capability to find relevant experimental literature. The framework supports multiple LLM providers (OpenAI, Anthropic, Google) through a common interface. In this study, we used GPT-5.1 for batch extraction and Claude Opus 4.6 for interactive extraction.

The prompt injects mechanistic context from the model structure: for each target parameter, the system retrieves the parameter's name, units, and description; the reactions where it appears with rate laws; the species involved; and the broader reaction network. This context guides the LLM toward experiments that constrain the target parameters. Source exclusion prevents reuse of papers across extractions to promote literature diversity. The prompt also includes the complete schema structure and guidance on common pitfalls (SEM vs. SD conversion, hardcoded constants, source relevance documentation), reducing the retry loop by preventing predictable errors.

## 2.2 Schema Design

The framework defines two complementary schemas for two calibration regimes. The *SubmodelTarget* schema captures isolated experiments (in vitro assays, single-mechanism studies) that constrain individual model parameters through simplified forward models. The *CalibrationTarget* schema captures clinical and in vivo endpoints that constrain the full model through observables computed over the complete species state. Both schemas share a common design philosophy but differ in how they connect literature data to model parameters.

### 2.2.1 Shared Design Principles

Both schemas enforce a data-first architecture: extraction of what the paper reports is separated from specification of how to use it for calibration. Every extracted numeric value must appear verbatim in a quoted text snippet from the source, creating an auditable trail and enabling automated hallucination detection. Both schemas require structured source documentation (DOI, title, authors), source relevance assessment (indication match, species translation, TME compatibility), and experimental context metadata (species, system type, disease stage). When YAML is parsed, the framework validates all constraints automatically, with structured error messages providing actionable feedback for the LLM retry loop.

### 2.2.2 SubmodelTarget: Parameter Priors from Isolated Experiments

The SubmodelTarget schema structures calibration targets into two layers: an *inputs* layer that captures data exactly as reported in the literature (value, units, uncertainty, sample size, source reference, and quoted snippet), and a *calibration* layer that specifies how that data constrains model parameters through a simplified forward model. For time-course data, this is typically a simplified ODE (e.g., exponential growth $dy/dt = k \cdot y$); for derived measurements, an algebraic model may suffice (e.g., $t_{1/2} = \ln(2)/k$). In both cases, the submodel derives from the full QSP model by isolating relevant species. Shared parameter names establish the connection: a proliferation rate named `k_apsc_prolif` in a submodel refers to the same parameter in the full model, allowing posteriors to serve directly as priors for full-model calibration. Each parameter includes a prior distribution with documented rationale, and an error model specifies the likelihood function connecting predictions to data. Supplementary Section 6 provides detailed field descriptions with YAML examples.

**Forward model type system.** The schema provides 15 built-in forward model types organized in four categories: ODE templates (exponential growth, first-order decay, two-state transitions, logistic growth, saturation, Michaelis-Menten), structured steady-state types for equilibrium measurements, a batch accumulation type for secretion assays, and generic fallbacks (algebraic, direct fit, custom). The complete type system is described in Supplementary Table 1. Each structured model declares its fields as typed roles (calibration parameter, extracted input, reference value, or literal) rather than raw values, allowing the same template to serve different estimation scenarios depending on which field is treated as the unknown.

A complete SubmodelTarget example is provided in Supplementary Section 1.

#### 2.2.3 CalibrationTarget: Clinical and In Vivo Endpoints

Many calibration constraints come from clinical or in vivo studies where the measurement reflects integrated model behavior. For example, intratumoral CD8+ T cell density reflects the net effect of recruitment, proliferation, exhaustion, and death. The CalibrationTarget schema represents these full-model endpoints through three components. The *observable* specifies how to compute the measurement from model species as a Python function that receives the solved state and returns a quantity with units; named constants carry provenance traced to a curated reference database or literature source. The *empirical data* block derives summary statistics (median and 95% CI) from literature inputs via Monte Carlo simulation, handling complications like subgroup mixing, SEM-to-SD conversion, and non-Gaussian uncertainty. For intervention studies, a *scenario* block captures treatment context (agents, doses, schedules). Complete examples and schema definitions are provided in Supplementary Sections 7 and 5.

### 2.2.4 Source Relevance Assessment

Experimental data rarely matches the target model perfectly. The `source_relevance` block documents the translation from source context to model context across several dimensions: indication match (exact, related, proxy, or unrelated), species translation (e.g., mouse to human), evidence type (clinical, in vitro, animal, review), and tumor microenvironment compatibility. Each dimension requires justification when the source deviates from the target model. An `estimated_translation_uncertaint` field quantifies the combined expected uncertainty from all translation factors, which propagates to prior width during inference (e.g., cross-species extraction from a proxy indication might warrant 10-fold uncertainty). Supplementary Section 13 provides detailed field descriptions.

### 2.2.5 Schema Implementation

Both schemas are implemented using Pydantic [22] with typed fields and constraints validated automatically during YAML parsing. Validation error messages are detailed and structured (e.g., reporting field path, expected type, and actual value), providing actionable feedback for the LLM retry loop. SubmodelTarget uses discriminated unions for its 15 forward model types; CalibrationTarget uses a class hierarchy with shared base models. Supplementary Section 5 provides the complete schema definitions and Supplementary Section 12 provides implementation details.

## 2.3 Validation Framework

Both schemas share a common validation core (DOI resolution, value-in-snippet matching, unit parsing, source relevance checks) with schema-specific extensions. Each validator runs independently, and a target must pass all validators before proceeding to downstream use.

### 2.3.1 Reference Validation

LLMs sometimes generate references to entities that do not exist elsewhere in the document. Reference validators check that every `uses_inputs` reference matches a defined input name, every `source_ref` matches a defined source, parameters referenced in the model exist in the parameter list, and observable definitions reference valid state variables.

### 2.3.2 Content Validation

Content validators address fabrication of values, citations, and other factual claims. *DOI validation* queries CrossRef for every DOI; title validation further checks that the extracted paper title fuzzy-matches the CrossRef metadata (75% similarity threshold), catching cases where a valid DOI is paired with fabricated bibliographic details. *Value-in-snippet validation* compares each extracted numeric value against its associated source text, providing the primary anti-hallucination defense: when an LLM hallucinates a number, it rarely fabricates a consistent snippet, and the mismatch provides a reliable detection signal. A separate external validator (Section 2.3.3) further verifies that snippets appear in the cited papers. *Unit validation* parses all unit strings using Pint [23] to verify dimensional validity. *Code validation* executes observation and observable code with mock data to verify correct behavior, catching logic errors and signature mismatches before inference. *Hardcoded constant detection* scans code blocks for undocumented numeric literals that should be traced to inputs or reference values. *Scale consistency validation* checks that observable outputs respond appropriately to varying inputs and have expected magnitudes. *Source relevance validators* enforce minimum uncertainty thresholds for cross-species (2-fold), cross-indication (3-fold), and low TME compatibility (10-fold) extractions, ensuring translation uncertainty propagates to prior widths.

Each validator raises a specific exception class with a `category` attribute (hallucination, fabrication, code, units, reference, structural, prior) that enables aggregation across extraction runs and feeds into the metrics reported in Section 3. Exception messages are designed for the LLM retry loop, including both the specific failure and guidance for correction. Optional Logfire instrumentation [24] provides OpenTelemetry-based tracing for debugging extraction failures. Supplementary Section 3 provides implementation details for each validator.

### 2.3.3 External Validation

An additional external validator verifies that value snippets actually appear in the cited source papers by fetching paper text from Europe PMC and Unpaywall using the DOI, then using fuzzy matching (80% similarity threshold) to locate each snippet. This catches a subtle failure mode where the LLM fabricates plausible-looking snippets that contain the correct value but do not actually appear in the cited paper. Because this requires network access to external APIs, it runs as a separate manual step.

#### 2.3.4 Structural and CalibrationTarget-Specific Validation

Structural validators check schema-level constraints (time span ordering, required fields per model type, array dimension consistency). CalibrationTarget adds validators for full-model observables: observable code execution against mock species data, a denominator audit for density/fraction observables (catching a common source of order-of-magnitude errors), species existence checks, and distribution code verification that computed summary statistics match reported empirical data within tolerance (1% for median, 10% for CI bounds). Supplementary Section 3 provides implementation details for all validators.

### 2.4 Code Generation

Validated SubmodelTarget specifications can be automatically translated into Julia scripts for Bayesian inference using Turing.jl [25]. The translator generates self-contained scripts with observed data constants, forward model functions, prior distributions, and likelihood terms. When multiple targets share parameters (identified by matching parameter names), the translator generates a joint model that declares each shared parameter once and constrains it from all relevant data sources simultaneously, yielding tighter posteriors than independent calibration. We chose to generate complete, readable scripts rather than provide a library abstraction, so that users can inspect, modify, and debug the inference model directly. Supplementary Section 4 provides details on the generated code structure, forward model handling, and inference diagnostics.

The two schemas connect through a natural calibration pipeline: SubmodelTarget posteriors from isolated experiments become informative priors for full-model calibration against CalibrationTarget endpoints. Because SubmodelTarget parameter names match the full model, the posterior distributions transfer directly without post-hoc mapping.

### 2.5 Application: PDAC QSP Model

The PDAC QSP model used in this study is adapted from a spatial QSP model of hepatocellular carcinoma [26] that first incorporated fibroblast dynamics into the tumor microenvironment. We evaluate MAPLE on this model using both schemas: 18 SubmodelTargets for individual parameter calibration from isolated experiments (spanning tumor growth, stromal dynamics, immune recruitment, and cytokine signaling), and 59 CalibrationTargets for full-model endpoints (baseline tumor microenvironment composition, neoadjuvant immunotherapy response, and clinical progression).

### 2.5.1 Evaluation Metrics

We report metrics across four workflow stages: extraction performance (pass rate, retries, time, tokens), error analysis (failures by category and tool usage), source characteristics (source types, species translation, indication match), and curation analysis (revision rates for batch vs. interactive modes, edit classification by content type).

# 3 Results

## 3.1 SubmodelTarget Extraction and Validation

We applied MAPLE to extract SubmodelTargets for a PDAC QSP model, covering tumor growth, stromal dynamics, immune recruitment, and cytokine signaling. An initial batch of 18 targets was extracted through the automated retry loop (Table 1); additional targets were extracted interactively during curation, yielding a final curated set of 37 SubmodelTargets used for joint inference. The metrics below characterize the automated first pass; Section 3.2 discusses the subsequent human curation step.

None of the extractions passed validation on the first attempt; all required at least one automated retry (Figure 1). Seven targets succeeded after a single retry, while parameters like CCL2 secretion and PSC activation required up to 7 iterations to resolve unit ambiguities or conflicting literature. Each extraction took 7.9 minutes on average, consuming 217,735 tokens per target (3.9M total). The validators caught 24 errors during extraction (Table 2), with unit errors most common, followed by fabrication (invalid DOIs), prior specification, and code errors. The LLM used 210 web searches and 22 code executions.

### 3.1.1 Source Characteristics

The 37 curated targets drew from diverse sources (Supplementary Section 8): 45% human clinical, with the remainder from animal studies. About 56% used human data directly; the rest

Table 1: Extraction metrics per calibration target.

| Target | Retries | Duration (min) | Tokens |
|---|---|---|---|
| N_IL2_CD4 | 3 | 6.5 | 198,267 |
| N_IL2_CD8 | 1 | 3.4 | 98,233 |
| initial_tumour_diameter | 1 | 3.0 | 70,219 |
| k_C1_growth | 1 | 4.1 | 81,606 |
| k_CCL2_sec | 5 | 9.2 | 367,754 |
| k_ECM_apsc_sec | 5 | 10.8 | 285,309 |
| k_ECM_deg | 1 | 5.8 | 135,317 |
| k_ECM_qpsc_sec | 5 | 12.0 | 406,891 |
| k_M1_pol | 1 | 7.1 | 190,149 |
| k_MDSC_rec | 3 | 8.4 | 221,048 |
| k_Mac_rec | 3 | 10.0 | 239,265 |
| k_apsc_death | 2 | 5.4 | 134,626 |
| k_psc_activation | 7 | 9.9 | 354,840 |
| k_psc_const | 2 | 7.7 | 209,064 |
| k_qpsc_death | 1 | 3.8 | 91,974 |
| n_CD8_clones | 1 | 5.3 | 114,839 |
| n_Treg_clones | 2 | 5.7 | 133,558 |
| q_Treg_T_in | 6 | 23.9 | 586,283 |
| **Average** | **2.8** | **7.9** | **217,735** |

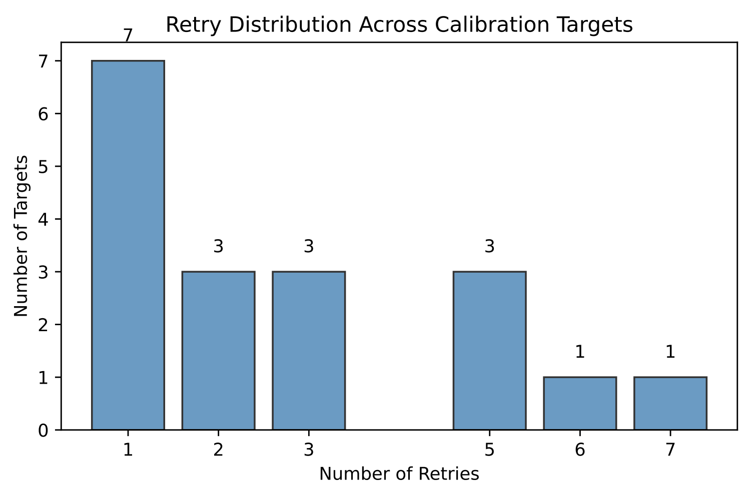


Figure 1: Distribution of automated retry counts across 18 extraction attempts.

Table 2: Error categories caught by validators during batch extraction. Units: incorrect or inconsistent unit conversions. Prior: proposed uncertainty range incompatible with stated translation context. Fabrication: LLM-generated DOIs or citations that do not resolve to real publications. Code: observation function fails unit checks or execution. Hallucination: extracted values not found in the cited source text. Each target may trigger multiple errors.

| Error Category | Count | % |
|---|---|---|
| units | 9 | 38% |
| prior | 5 | 21% |
| fabrication | 4 | 17% |
| code | 4 | 17% |
| other | 1 | 4% |
| hallucination | 1 | 4% |
| **Total** | **24** | |

required cross-species scaling from mouse (35%) or rat (8%). Only 45% came from PDAC-specific experiments; most (43%) relied on proxy indications.

### 3.2 Human Curation

To quantify human curation, we compared raw LLM output against final curated versions. All curation was performed interactively with Claude Code (claude-opus-4-6), with the modeler directing edits and modeling decisions while the LLM handled text generation, code writing, and literature comprehension. No batch extraction was used without modification; in a documented review of one batch (12 CalibrationTarget files), 58% were rejected outright.

For the 37 batch-extracted SubmodelTarget files, the modeler changed the forward model type in 65% of files (24/37), adjusted prior distribution parameters in 46%, changed input data values in 46%, and revised source relevance assessments in every file. For the 22 batch-extracted CalibrationTargets, the modeler revised observable code in 32% of files, changed species lists in 23%, and adjusted empirical data in 18%.

The remaining 27 CalibrationTargets were extracted interactively with Claude Code using source PDFs loaded directly into the session. The modeler provided domain guidance (species mapping, denominator definitions, distribution choices) while the LLM handled literature comprehension and code generation. These files required minimal subsequent revision (a single shared reference value correction across 7,483 total lines); modeler judgment was embedded in the extraction process rather than applied post-hoc.

## 3.3 CalibrationTarget Extraction

We applied the CalibrationTarget schema to 59 full-model endpoints for the same PDAC QSP model, organized into three scenario groups: 35 baseline (no treatment) targets capturing the untreated tumor microenvironment, 18 neoadjuvant immunotherapy targets (GVAX ± nivolumab), and 1 clinical progression target (Table 3). All 59 targets pass schema validation including observable code execution, distribution code verification, and unit consistency checks.

Table 3: CalibrationTarget summary by scenario group.

| Scenario Group | Targets |
|---|---|
| Baseline (no treatment) | 35 |
| Neoadjuvant immunotherapy (GVAX ± nivolumab) | 18 |
| Clinical progression | 1 |
| **Total** | **59** |

Of the 59 targets, 22 were extracted by GPT-5.1 through the batch pipeline, 27 interactively with Claude Opus 4 using source PDFs, and 10 manually. The 18 neoadjuvant immunotherapy targets include per-patient data digitized from scatter plots (25% of all CalibrationTargets). Curation metrics for both extraction modes are reported in Section 3.2.

The observables span 25 unique model species, with complexity ranging from 1 to 19 species per target (mean 7.2). The targets drew from 25 unique primary sources (from years 2000–2026). Supplementary Section 9 provides detailed per-target metrics.

## 3.4 Code Generation and Inference

All 18 SubmodelTargets were translated to a joint Julia inference script targeting Turing.jl [25]. The translator identified 19 unique parameters and generated corresponding priors and likelihood terms. All parameters converged ($\hat{R} < 1.01$, ESS $> 3000$), with posterior estimates yielding biologically plausible values (e.g., tumor doubling time of $\sim$130 days, consistent with clinical PDAC observations). Full inference results including convergence diagnostics, posterior distributions, and model type breakdown are reported in Supplementary Section 10.

# 4 Discussion

MAPLE demonstrates that structured schemas can serve as a productive collaboration interface between LLMs and modelers for QSP model calibration. The curation analysis reveals that modeler judgment contributes roughly half of the final calibration target content. Rather than automating away the modeler, the framework restructures the collaboration: the schema defines what must be captured, validators catch errors from both LLM and human sources, and the separation of data extraction from modeling decisions clarifies where different types of expertise are needed.

## 4.1 Validation as the Core Mechanism

None of the 18 SubmodelTarget extractions passed validation on the first attempt, and the same automated retry pipeline was used for batch-extracted CalibrationTargets (detailed retry metrics are reported for SubmodelTargets, where the Logfire-instrumented pipeline provided per-target tracing). This is expected behavior: the validators serve as an automated first-pass review, and the retry loop resolves most issues without human intervention. Extractions requiring multiple retries (up to 7) typically involved ambiguous unit conventions or conflicting literature values.

The value-in-snippet requirement proved particularly effective, catching a common LLM behavior: paraphrasing source text rather than quoting it exactly. The validator flags these cases, and the retry loop produces verbatim snippets. A separate external validator checks that snippets actually appear in the cited papers, catching cases where the LLM generates plausible text that does not exist in the source. Together, these validators support human review: a reviewer can verify each extraction by checking the snippet against the source paper, without re-reading entire methods sections.

## 4.2 Schema Design Choices

The separation of data extraction from model specification clarifies where human judgment is required (the inputs should be objective; the calibration layer involves modeling choices), enables reuse (if a modeling assumption changes, the inputs remain valid), and supports independent review of data versus modeling decisions.

The dual-schema architecture applies the same data-first principles at two scales of calibration.

SubmodelTarget operates on isolated experiments that constrain individual parameters through simplified forward models; CalibrationTarget operates on clinical and in vivo endpoints that constrain the integrated model through full-species observables. Shared design elements (source provenance, relevance assessment, experimental context, value-snippet validation) ensure consistent rigor across both scales, while schema-specific components (forward model types vs. observable code, error models vs. distribution code) capture the distinct requirements of each calibration regime.

### 4.3 Modeler Contribution and Limitations of Extracted Data

The curation results in Section 3.2 confirm that MAPLE does not reduce the need for modeler expertise: for SubmodelTargets, the modeler changed forward model types, adjusted prior distributions, revised input data, and rewrote source relevance assessments across the majority of batch-extracted files; for CalibrationTargets, the modeler revised observable code, corrected species lists, and adjusted empirical data values. These are scientific decisions, not formatting corrections. The framework restructures, rather than replaces, this expertise by separating tasks with different skill requirements. The structured schema ensures documentation completeness: every input requires a source reference and snippet, every prior requires a rationale, every cross-indication extraction requires a justification. The two collaboration modes (batch pipeline vs. interactive extraction) offer different trade-offs (Supplementary Section 14).

The source characteristics reveal a fundamental challenge in QSP calibration: only 45% of SubmodelTargets came from PDAC-specific experiments, and 35% required cross-species translation from mouse data. The schema addresses this through the `source_relevance` block, which requires that translation uncertainty propagate to prior widths. A parameter calibrated from mouse fibrosis data with 5-fold translation uncertainty will have a correspondingly wide posterior, appropriately reflecting the limited applicability of available evidence.

### 4.4 Relationship to Existing Approaches

MAPLE addresses a gap between general-purpose biomedical extraction frameworks [17, 18], which do not address QSP-specific downstream requirements (uncertainty quantification, forward model specification, code generation), and QSP-focused tools like QSP-Copilot [16], which emphasize model structure discovery rather than quantitative parameter calibration with provenance. The validation approach combines grounding strategies from retrieval-augmented generation [27] and constrained decoding [28] with domain-specific validators (DOI resolution, unit parsing, code execution) tailored to quantitative scientific extraction.

Unlike knowledge graph approaches that prioritize coverage across thousands of abstracts [14, 29] and can tolerate some extraction errors, parameter calibration requires higher per-target stringency because each value directly influences model predictions. MAPLE aligns with IQ Consortium best practices for AI/ML in quantitative modeling [7], emphasizing explainability, human-in-the-loop oversight, and documented provenance. While evaluated on a QSP model, the approach applies to any parameter-rich mechanistic model that draws calibration data from literature. A more detailed comparison of individual tools and approaches is provided in Supplementary Section 15.

### 4.5 Limitations and Future Work

Several limitations merit discussion. First, the framework relies on the LLM's built-in web search for source discovery, so extraction quality is bounded by the LLM's ability to locate and access papers. Closed-access publications may be only partially available (e.g., abstract only), limiting extraction depth. Emerging agentic capabilities (LLM-controlled browsers with institutional authentication) may address this. The framework also depends on evolving LLM capabilities: extraction is stochastic, and performance characteristics vary across providers and model versions.

Second, modeler intervention is pervasive. For SubmodelTargets, the modeler changed the forward model type in 65% of files and adjusted prior parameters in 46%. Prompt improvements could reduce some recurring errors, but core scientific decisions will remain modeler responsibilities regardless of LLM capability. CalibrationTargets required different curation: observable code (changed in 32% of batch-extracted files), Monte Carlo distribution derivation, and denominator audit fields demand modeling expertise that current LLMs provide as a starting point. The schema and validators can also be used without LLMs, providing structure for purely manual curation.

Third, extracting structured quantitative data from figures remains challenging; the neoadjuvant immunotherapy targets required manual digitization of scatter plots. Integrating multimodal LLM vision capabilities is a natural direction for future work. Finally, this evaluation covers a single model (PDAC QSP); broader evaluation across therapeutic areas would better characterize generalizability.

# Study Highlights

**What is the current knowledge on the topic?**

QSP model calibration requires data from published literature, spanning experimental studies and clinical endpoints. Manual curation is labor-intensive with inconsistent documentation, while LLM extraction exhibits hallucination and fabrication errors unacceptable for quantitative modeling.

**What question did this study address?**

Can structured schemas serve as a collaboration interface between LLMs and modelers for QSP model calibration, and what is the relative contribution of each?

**What does this study add to our knowledge?**

Structured validation schemas enable LLM-modeler collaboration through batch or interactive extraction. Both modes required substantial modeler input, with forward model types changed in 65% of SubmodelTargets and observable code revised in 32% of CalibrationTargets. Targeted validators catch characteristic LLM errors, and code generation enables joint Bayesian inference.

**How might this change drug discovery, development, and/or therapeutics?**

The framework produces reproducible, provenance-rich calibration targets that improve documentation consistency across modeling teams while making modeler contributions explicit and auditable.

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# Author Contributions

J.E. designed research, performed research, analyzed data, and wrote the manuscript. A.S.P. designed research and wrote the manuscript.

# Data Availability Statement

MAPLE (the SubmodelTarget and CalibrationTarget schemas, validation framework, and Julia code generator) is available at `https://github.com/popellab/maple`. Extraction inputs, curated YAML targets, and scripts to reproduce all paper statistics are available at `https://github.com/popellab/maple-paper`. The PDAC calibration targets used in this study (18 SubmodelTargets and 59 CalibrationTargets) are provided as supplementary YAML files. Generated Julia inference scripts are included in the supplementary materials.

# References


[1] K Gadkar, DC Kirouac, DE Mager, PH van der Graaf, and S Ramanujan. A Six-Stage Workflow for Robust Application of Systems Pharmacology. *CPT: Pharmacometrics & Systems Pharmacology*, 5(5):235–249, May 2016. ISSN 2163-8306. doi: 10.1002/psp4.12071.

[2] Hanwen Wang, Theinmozhi Arulraj, Alberto Ippolito, and Aleksander S. Popel. Quantitative Systems Pharmacology Modeling in Immuno-Oncology: Hypothesis Testing, Dose Optimization, and Efficacy Prediction. *Handbook of Experimental Pharmacology*, 289:261–284, 2025. ISSN 0171-2004. doi: 10.1007/164_2024_735.

[3] Hanwen Wang, Theinmozhi Arulraj, Alberto Ippolito, and Aleksander S. Popel. From virtual patients to digital twins in immuno-oncology: Lessons learned from mechanistic quantitative systems pharmacology modeling. *npj Digital Medicine*, 7(1):189, July 2024. ISSN 2398-6352. doi: 10.1038/s41746-024-01188-4.

[4] Theodore R. Rieger, Richard J. Allen, Lukas Bystricky, Yuzhou Chen, Glen Wright Colopy, Yifan Cui, Angelica Gonzalez, Yifei Liu, R. D. White, R. A. Everett, H. T. Banks, and Cynthia J. Musante. Improving the generation and selection of virtual populations in quantitative systems pharmacology models. *Progress in Biophysics and Molecular Biology*, 139:15–22, November 2018. ISSN 0079-6107. doi: 10.1016/j.pbiomolbio.2018.06.002.

[5] Morgan Craig, Jana L. Gevertz, Irina Kareva, and Kathleen P. Wilkie. A practical guide for

the generation of model-based virtual clinical trials. *Frontiers in Systems Biology*, 3, June 2023. ISSN 2674-0702. doi: 10.3389/fsysb.2023.1174647.

[6] Núria Folguera-Blasco, Florencia A. T. Boshier, Aydar Uatay, Cesar Pichardo-Almarza, Massimo Lai, Jacopo Biasetti, Richard Dearden, Megan Gibbs, and Holly Kimko. Coupling quantitative systems pharmacology modelling to machine learning and artificial intelligence for drug development: Its pAIns and gAIns. *Frontiers in Systems Biology*, 4, July 2024. ISSN 2674-0702. doi: 10.3389/fsysb.2024.1380685.

[7] Nadia Terranova, Didier Renard, Mohamed H. Shahin, Sujatha Menon, Youfang Cao, Cornelis E.C.A. Hop, Sean Hayes, Kumpal Madrasi, Sven Stodtmann, Thomas Tensfeldt, Pavan Vaddady, Nicholas Ellinwood, and James Lu. Artificial Intelligence for Quantitative Modeling in Drug Discovery and Development: An Innovation and Quality Consortium Perspective on Use Cases and Best Practices. *Clinical Pharmacology & Therapeutics*, 115(4):658–672, 2024. ISSN 1532-6535. doi: 10.1002/cpt.3053.

[8] Igor Goryanin, Irina Goryanin, and Oleg Demin. Revolutionizing drug discovery: Integrating artificial intelligence with quantitative systems pharmacology. *Drug Discovery Today*, 30(9): 104448, September 2025. ISSN 1878-5832. doi: 10.1016/j.drudis.2025.104448.

[9] Ioannis P. Androulakis, Lourdes Cucurull-Sanchez, Anna Kondic, Krina Mehta, Cesar Pichardo, Meghan Pryor, and Marissa Renardy. The dawn of a new era: Can machine learning and large language models reshape QSP modeling? *Journal of Pharmacokinetics and Pharmacodynamics*, 52(4):36, June 2025. ISSN 1573-8744. doi: 10.1007/s10928-025-09984-5.

[10] Ioannis P. Androulakis, Limei Cheng, Carolyn R. Cho, and Tongli Zhang. Leveraging large language models to compare perspectives on integrating QSP and AI/ML. *Journal of Pharmacokinetics and Pharmacodynamics*, 52(3):29, May 2025. ISSN 1573-8744. doi: 10.1007/s10928-025-09976-5.

[11] Sendong Zhao, Chang Su, Zhiyong Lu, and Fei Wang. Recent advances in biomedical literature mining. *Briefings in Bioinformatics*, 22(3):bbaa057, May 2021. ISSN 1477-4054. doi: 10.1093/bib/bbaa057.

[12] Jiacheng Hu, Runyuan Bao, Yang Lin, Hanchao Zhang, and Yanlin Xiang. Accurate Medical Named Entity Recognition Through Specialized NLP Models. *arXiv preprint arXiv:2412.08255*, December 2024. doi: 10.48550/arXiv.2412.08255.

[13] Ferran Gonzalez Hernandez, Quang Nguyen, Victoria C. Smith, José Antonio Cordero, Maria Rosa Ballester, Màrius Duran, Albert Solé, Palang Chotsiri, Thanaporn Wattanakul, Gill Mundin, Watjana Lilaonitkul, Joseph F. Standing, and Frank Kloprogge. Named entity recognition of pharmacokinetic parameters in the scientific literature. *Scientific Reports*, 14 (1):23485, October 2024. ISSN 2045-2322. doi: 10.1038/s41598-024-73338-3.

[14] Shan He, Yukun Tan, Qing Ye, Matthew M Gubin, Merve Dede, Hind Rafei, Weiyi Peng, Katayoun Rezvani, Vakul Mohanty, and Ken Chen. AI-powered Immune Cell Knowledge Graph (ICKG) with granular immune contexts enables immune program interpretation. *npj Artificial Intelligence*, 2(1):13, January 2026. ISSN 3005-1460. doi: 10.1038/s44387-025-00060-4.

[15] Muhammad Ali Khan, Umair Ayub, Syed Arsalan Ahmed Naqvi, Kaneez Zahra Rubab Khakwani, Zaryab Bin Riaz Sipra, Ammad Raina, Sihan Zou, Huan He, Seyyed Amir Hossein, Bashar Hasan, R. Bryan Rumble, Danielle S. Bitterman, Jeremy L. Warner, Jia Zou, Amye J. Tevaarwerk, Konstantinos Leventakos, Kenneth L. Kehl, Jeanne M. Palmer, M. Hassan Murad, Chitta Baral, and Irbaz Bin Riaz. Collaborative Large Language Models for Automated Data Extraction in Living Systematic Reviews. *medRxiv: The Preprint Server for Health Sciences*, page 2024.09.20.24314108, September 2024. doi: 10.1101/2024.09.20.24314108.

[16] Anuraag Saini and Ali Farnoud. QSP-Copilot: An AI-Augmented Platform for Accelerating Quantitative Systems Pharmacology Model Development. *CPT: Pharmacometrics & Systems Pharmacology*, 14(11):1775–1786, 2025. ISSN 2163-8306. doi: 10.1002/psp4.70127.

[17] Enshuo Hsu and Kirk Roberts. LLM-IE: A python package for biomedical generative information extraction with large language models. *JAMIA Open*, 8(2):ooaf012, April 2025. ISSN 2574-2531. doi: 10.1093/jamiaopen/ooaf012.

[18] Xingbo Wang, Samantha L. Huey, Rui Sheng, Saurabh Mehta, and Fei Wang. SciDaSynth: In-

teractive Structured Data Extraction From Scientific Literature With Large Language Model. *Campbell Systematic Reviews*, 21(4):e70073, 2025. ISSN 1891-1803. doi: 10.1002/cl2.70073.

[19] Sebastian Farquhar, Jannik Kossen, Lorenz Kuhn, and Yarin Gal. Detecting hallucinations in large language models using semantic entropy. *Nature*, 630(8017):625–630, June 2024. ISSN 1476-4687. doi: 10.1038/s41586-024-07421-0.

[20] Lei Huang, Weijiang Yu, Weitao Ma, Weihong Zhong, Zhangyin Feng, Haotian Wang, Qianglong Chen, Weihua Peng, Xiaocheng Feng, Bing Qin, and Ting Liu. A Survey on Hallucination in Large Language Models: Principles, Taxonomy, Challenges, and Open Questions. *ACM Transactions on Information Systems*, 43(2):1–55, March 2025. ISSN 1046-8188, 1558-2868. doi: 10.1145/3703155.

[21] Shrey Pandit, Jiawei Xu, Junyuan Hong, Zhangyang Wang, Tianlong Chen, Kaidi Xu, and Ying Ding. MedHallu: A Comprehensive Benchmark for Detecting Medical Hallucinations in Large Language Models, February 2025.

[22] Samuel Colvin, Eric Jolibois, Hasan Ramezani, Adrian Garcia Badaracco, Terrence Dorsey, David Montague, Serge Matveenko, Marcelo Trylesinski, Sydney Runkle, David Hewitt, Alex Hall, and Victorien Plot. Pydantic Validation, October 2025. URL `https://github.com/pydantic/pydantic`.

[23] Hernán E. Grecco. Pint: Operate and manipulate physical quantities in Python, 2025. URL `https://github.com/hgrecco/pint`.

[24] Pydantic. Logfire: Observability platform for Python applications, 2025. URL `https://pydantic.dev/logfire`.

[25] Hong Ge, Kai Xu, and Zoubin Ghahramani. Turing: A Language for Flexible Probabilistic Inference. In *Proceedings of the Twenty-First International Conference on Artificial Intelligence and Statistics*, pages 1682–1690. PMLR, March 2018. URL `https://proceedings.mlr.press/v84/ge18b.html`.

[26] Shuming Zhang, Hanwen Wang, Yeonju Cho, Wendy Wong, Mark Yarchoan, Elizabeth M. Jaffee, Won Jin Ho, Luciane T. Kagohara, Elana J. Fertig, Aleksander S. Popel, and Atul

Deshpande. Quantitative Calibration of a Spatial QSP Model Identifies Fibroblast Impact on HCC Immunotherapy. *bioRxiv*, page 2025.10.31.685939, November 2025. ISSN 2692-8205. doi: 10.1101/2025.10.31.685939.

[27] Yihan Li, Xiyuan Fu, Ghanshyam Verma, Paul Buitelaar, and Mingming Liu. Mitigating Hallucination in Large Language Models (LLMs): An Application-Oriented Survey on RAG, Reasoning, and Agentic Systems, October 2025.

[28] Saibo Geng, Hudson Cooper, Michał Moskal, Samuel Jenkins, Julian Berman, Nathan Ranchin, Robert West, Eric Horvitz, and Harsha Nori. JSONSchemaBench: A Rigorous Benchmark of Structured Outputs for Language Models. *arXiv preprint arXiv:2501.10868*, February 2025. doi: 10.48550/arXiv.2501.10868.

[29] Ting Gao, Xue Zhai, Chuan Yang, Linlin Lv, and Han Wang. Joint extraction of entity and relation based on fine-tuning BERT for long biomedical literatures. *Bioinformatics Advances*, 4(1):vbae194, January 2024. ISSN 2635-0041. doi: 10.1093/bioadv/vbae194.